



**Microsoft**  
CERTIFIED

**AZURE AI  
FUNDAMENTALS**



## **Identify AI concepts and capabilities (40–45%)**

### **Describe principles of responsible AI**

- Describe considerations for fairness in an AI solution
- Describe considerations for reliability and safety in an AI solution
- Describe considerations for privacy and security in an AI solution
- Describe considerations for inclusiveness in an AI solution
- Describe considerations for transparency in an AI solution
- Describe considerations for accountability in an AI solution

### **Identify AI model components and configurations**

- Describe how generative AI models work
- Identify an appropriate AI model, based on capabilities
- Identify appropriate model deployment options and configuration parameters

### **Identify AI workloads**

- Identify scenarios for common AI workloads, including generative and agentic AI, text analysis, speech, computer vision, and information extraction
- Describe common text analysis techniques, including keyword extraction, entity detection, sentiment analysis, and summarization
- Identify features and capabilities of speech recognition and speech synthesis
- Identify features and capabilities of computer vision and image-generation models
- Identify techniques to extract information from text, images, audio, and videos

## **Implement AI solutions by using Microsoft Foundry (55–60%)**

### **Implement generative AI apps and agents by using Foundry**

- Create effective system and user prompts for generative AI models
- Deploy a model and interact with it in the Foundry portal
- Create a lightweight chat client application by using the Foundry SDK
- Create and test a single-agent solution in the Foundry portal
- Create a lightweight client application for an agent

### **Implement AI solutions for text and speech by using Foundry**

- Build a lightweight application that includes text analysis
- Respond to spoken prompts by using a deployed multimodal model
- Build a lightweight application by using Azure Speech in Foundry Tools

### **Implement AI solutions with computer vision and image-generation capabilities by using Foundry**

- Interpret visual input in prompts by using a deployed multimodal model
- Create new visual outputs by using generative models
- Build a lightweight application that includes vision capabilities

### **Implement AI solutions for information extraction by using Foundry**

- Extract information from documents and forms by using Azure Content Understanding in Foundry Tools
- Extract information from images by using Content Understanding
- Extract information from audio and video by using Content Understanding
- Build a lightweight application with information extraction capabilities by using Content Understanding

# Introduction to Artificial Intelligence (AI)

- **Artificial Intelligence (AI):**
  - Computer systems performing tasks that normally require human intelligence
  - Examples: visual perception, speech recognition, decision making, language translation
  - Represents the concept of human intelligence inside machines

## Types of Artificial Intelligence

- **General AI:**
  - Self-aware systems that can teach themselves new tasks
  - Exists mainly in science fiction, not in real life
- **Narrow AI:**
  - Systems with human-like intelligence limited to specific tasks
  - Examples: voice assistants, smart speakers, scheduling, weather queries
  - Common in daily life

## Subfields of Artificial Intelligence

- **Machine Learning (ML):**
  - Computers learn to develop their own algorithms
  - **Supervised Learning:**
    - Data is labeled and instructions are provided
  - **Unsupervised Learning:**
    - Computer finds patterns in data without explicit instructions
- **Natural Language Processing (NLP):**
  - Machines read, understand, and translate human languages
  - Applications: language translation, question answering, sentiment analysis
- **Perception:**
  - Use of input from images, video, audio, lidar, sonar
  - Applications: facial recognition, speech recognition, object recognition

## Real-World Applications of AI

- Self-driving cars and driver assistance features
- Smart home devices that learn user preferences
- Home assistants for conversation, music, and weather
- Real-time transcription for accessibility
- AI coaching in customer service calls

## Ethical Considerations and Challenges

- Privacy concerns from cameras and sensors collecting data
- Profiling based on age, gender, or other characteristics without user knowledge

## Core Focus: The Machine Learning Model

- The model is central to all machine learning tasks
- Steps include developing, testing, deploying, and maintaining the model
- A good model can outperform traditional programming in certain tasks
- A poor model will underperform and is not desirable

## What is a Machine Learning Model?

- Used mainly as a prediction engine to forecast future outcomes
- Unlike traditional programming, models learn from data instead of following hard-coded rules
- Example:
  - Traditional traffic lights follow fixed, human-programmed patterns
  - Machine learning models analyze traffic data to optimize light changes for better flow

## Pattern Recognition and Prediction

- Models recognize patterns in data, such as traffic trends at different times of day
- Can predict future behaviors based on learned patterns
- Subject matter experts can spot patterns, but models can analyze large amounts of data more efficiently

## Types of Machine Learning Algorithms

- **Regression:** Predicts future values (e.g., traffic volume for tomorrow)
- **Categorization:** Classifies items (e.g., identifying objects in photos, recognizing cities or landmarks)
- **Recognition:** Identifies people, celebrities, or landmarks in images
- **Speech and Text Models:** Understand spoken or written language and user intent

## Model Training and Evaluation

- **Training:** Feeding data to the model so it can learn patterns
- **Evaluation:** Testing the model to measure accuracy and precision
  - Some models have binary outcomes (right or wrong)
  - Others provide percentage-based predictions (how close the guess is)
- Evaluation methods help determine if the model is good enough for use

## Deployment and Maintenance

- After training and evaluation, the model is deployed for real-world use
- Ongoing maintenance ensures the model continues to perform well over time

## Principles of Responsible AI

### Why Responsible AI Matters

- AI is widely used in society: work, home, government, retail, and more
- Ethical and moral challenges arise when decisions are made by computers
- AI can make mistakes, be unexplainable, or have unintended consequences
- Laws and regulations may restrict how AI can use sensitive data
- Examples of unintended consequences:
  - Facial recognition not working equally for all skin tones
  - AI recruiting tools showing bias toward certain groups
  - Healthcare, banking, and law enforcement decisions made without human review
  - Privacy risks from data collection and leaks

## Microsoft's Six Principles of Responsible AI

- **Fairness**
  - AI systems should treat everyone equitably
  - Avoid bias based on race, gender, age, or other factors
  - Example: Loan or healthcare decisions should not favor one group over another
- **Reliability and Safety**
  - AI must operate reliably and safely in all conditions, including unexpected ones
  - Systems should handle errors and attacks gracefully
  - Example: Self-driving cars must work in bad weather and resist attempts to trick them
- **Privacy and Security**
  - AI must protect personal data and be secure against attacks
  - Respect privacy laws and user consent
  - Example: Do not use or share sensitive data without permission
- **Inclusiveness**
  - AI should benefit everyone and address a wide range of needs
  - Design systems for people with different abilities, languages, and backgrounds
  - Example: Voice assistants should work for people with accents or speech differences
- **Transparency**
  - People should understand how AI decisions are made
  - Be able to explain decisions, especially for high-impact outcomes
  - Example: Explain why a loan or job application was accepted or rejected
- **Accountability**
  - Developers and organizations are responsible for AI system outcomes
  - Humans must monitor, review, and correct AI decisions as needed
  - Example: Regularly check and update models to ensure they remain fair and accurate

## Considerations for Fairness in AI Solutions

### Definition of Fairness

- AI systems should produce equitable outcomes for all people
- No discrimination based on race, gender, age, disability, religion, or economic status
- Many countries have laws protecting against such discrimination

### Key Considerations for Fairness

- **Training Data Bias**
  - AI learns from historical data
  - If past decisions were biased, the AI will inherit those biases
  - Example: If more men were hired for IT roles in the past, the AI may continue this trend
- **Performance Parity**
  - Evaluate model accuracy across different demographic groups
  - Ensure the model performs equally well for all races, genders, and ages
  - Avoid models that work well for one group but poorly for another
- **Sensitive Feature Leakage**
  - Removing direct features (like race or gender) may not be enough
  - Indirect features (like university attended) can reveal sensitive information
  - AI may still infer and act on these hidden biases

- **Allocation vs. Quality of Service**
  - Fairness is not just about who gets selected, but also about the benefits received
  - Example: Equal pay, raises, or promotions for similar roles and performance
  - Ensure benefits and opportunities are distributed fairly
- **Feedback Loops**
  - Using AI-generated outcomes as new training data can reinforce and amplify existing biases
  - Monitor and update models to prevent unfair patterns from repeating over time

## Considerations for Reliability and Safety in AI Solutions

### Definition of Reliability and Safety

- AI systems must behave consistently, handle unexpected inputs gracefully, and avoid causing harm through failure, misuse, or attack

### Key Considerations for Reliability and Safety

- **Edge Cases**
  - Test AI systems with a wide range of conditions, not just typical scenarios
  - Ensure the model can handle unusual or rare situations in the real world
- **Failure Modes**
  - Understand how the model fails, not just if it fails
  - Avoid silent or confidently wrong failures; clear error messages are safer
- **Adversarial Robustness**
  - Anticipate and test for adversarial attacks (inputs designed to trick the model)
  - Use red-team testing, especially in high-stakes domains like finance or healthcare
- **Human Override**
  - Allow humans to override AI decisions when necessary
  - Important for critical systems like self-driving cars, bank loans, and healthcare
- **Deployment Drift (Degradation Under Distribution Shift)**
  - Monitor models after deployment; performance can degrade as real-world conditions change
  - Regularly update and retrain models to maintain accuracy and safety
- **Comprehensive Testing**
  - Go beyond accuracy metrics; include stress testing, load testing, and error handling
  - Test for various operational scenarios, including system restarts and long-term use
- **Safety Boundaries (Guardrails)**
  - Implement guardrails to prevent AI from generating harmful or dangerous content
  - Especially important for generative AI systems (text, images, videos)

## Considerations for Privacy and Security in AI Solutions

### Definition of Privacy and Security

- AI systems must protect personal and sensitive data, comply with privacy regulations, and be resilient against attacks that could compromise data or model integrity

## Key Considerations for Privacy and Security

- **Data Minimization**
  - Collect only the data strictly required for the task
  - Do not retain data longer than necessary
  - Avoid collecting unnecessary details (e.g., only store the number of credit cards, not card numbers)
- **Consent and Transparency**
  - Inform users about what data is collected and how it will be used
  - Obtain user consent and provide options to opt out or request deletion
  - Follow regulations like GDPR for user rights
- **Training Data Privacy**
  - Prevent models from reproducing sensitive or personal information from training data
  - Be aware that models can sometimes output real names or details if not properly managed
- **Model Inversion/Extraction Attacks**
  - Protect against attackers probing the model to reconstruct training data or extract model weights
  - Keep model weights confidential to prevent reverse engineering
- **Secure Infrastructure**
  - Secure endpoints, storage, and data pipelines
  - Use encryption for data in transit and at rest
  - Apply the principle of least privilege for access control
- **Synthetic Data and Anonymization**
  - Use anonymized or synthetic data for training when possible
  - Remove or mask personally identifiable information to protect privacy
- **Regulatory Compliance**
  - Ensure compliance with relevant laws and standards (e.g., HIPAA, PCI DSS, GDPR, EU AI Act)
  - Understand and meet obligations for handling data in AI models

## Considerations for Inclusiveness in AI Solutions

### Definition of Inclusiveness

- AI should be designed so everyone, including people with disabilities, those in low-resource settings, and marginalized communities, can access and benefit from it

### Key Considerations for Inclusiveness

- **Accessibility in User Interfaces**
  - Ensure UIs, voice interfaces, and outputs meet accessibility standards (e.g., WCAG 2)
  - Support users with visual, auditory, motor, and cognitive disabilities
  - Use features like alternative text, captions, and keyboard navigation
- **Language and Culture**
  - Design for multilingual and cross-cultural use
  - Avoid training only on English or a single culture's data
  - Plan for translation and localization to serve diverse users

- **Technology and Connectivity Access**
  - Consider users with limited internet or older devices
  - Offer low-bandwidth options, compressed images, text-only modes, or offline features
- **Representation in Training Data**
  - Use diverse training data that reflects the full range of humanity
  - Avoid bias by including images, voices, and text from different groups
- **Participatory Design**
  - Involve people from various backgrounds in the design process
  - Gather input from groups who may be affected by the AI system
- **Inclusive Defaults and Examples**
  - Avoid default settings or examples that favor one group (e.g., only male, western, or white-centered content)
  - Ensure default content and settings reflect a broad range of people

## Considerations for Transparency in AI Solutions

### Definition of Transparency

- People interacting with or affected by an AI system should understand how it works, why it produces certain outputs, and what its limitations are

### Key Considerations for Transparency

- **Explainability**
  - Provide clear, human-understandable explanations for how decisions are made
  - Offer both general (global) and specific (local) explanations for outputs
- **Model Cards and Documentation**
  - Supply detailed documentation about the AI model
  - Include information on training data, capabilities, intended uses, and limitations
- **Confidence and Uncertainty**
  - AI should express uncertainty when it is unsure about an answer
  - Avoid presenting uncertain outputs as facts
- **AI Identification**
  - Clearly inform users when they are interacting with an AI system
  - Prevent users from being misled into thinking they are communicating with a human
- **Logging and Auditability**
  - Log decisions, model versions, and key factors used in decision-making
  - Enable review and auditing, especially for high-impact systems
- **Stating Limitations**
  - Clearly communicate known limitations, failure modes, and out-of-scope areas
  - Indicate the date of last training or data coverage when relevant

## Considerations for Accountability in AI Solutions

### Definition of Accountability

- Humans and organizations that design, build, and deploy AI systems must remain responsible for how those systems behave
- AI decisions cannot be used as a shield against responsibility



## Key Considerations for Accountability

- **Named Ownership**
  - Every AI system in production must have a named owner or responsible team
  - Clear responsibility for outcomes and behavior of the AI system
- **Human Review and Override**
  - For high-stakes decisions (medical, legal, financial, safety), humans must be able to review, override, or escalate AI decisions
  - AI should support human judgment, not replace it
- **Governance and Oversight**
  - Establish processes and procedures for reviewing and approving AI systems
  - Use ethics boards, AI teams, and impact assessments before deployment
- **Audit Trails**
  - Maintain logs of decisions, model versions, training data, and approvals
  - Enable tracing and investigation of how and why decisions were made
- **Incident Response**
  - Have clear plans for identifying, investigating, and addressing failures or incidents
  - Teams should be ready to analyze and respond to issues
- **Regulatory Compliance**
  - Ensure compliance with relevant laws and regulations (e.g., EU AI Act, US executive orders)
  - Meeting legal obligations is mandatory
- **Vendor Accountability**
  - Remain accountable for third-party or vendor AI systems used
  - Responsibility to customers cannot be passed to vendors; deployers must answer for outcomes

## How Generative AI Models Work

### Definition of Generative AI

- Generative AI models create new content, such as text, images, code, or video
- Unlike traditional AI, which analyzes or classifies data, generative AI produces original outputs

### Key Concepts and Components

- **Deep Learning and Neural Networks**
  - Generative AI is built on deep learning, using neural networks with many layers
  - Neural networks learn patterns from large datasets through multiple training passes
- **Massive Training Data**
  - Models are trained on huge amounts of data (webpages, books, images, etc.)
  - More data leads to better and more generalizable models
- **Foundation Models**
  - Large, general-purpose models trained on broad data
  - Can be adapted or fine-tuned for specific domains (e.g., legal, medical)
- **Transformer Architecture**
  - A breakthrough design that allows models to process data in parallel
  - Uses "attention" to focus on relevant parts of input, improving understanding and efficiency
  - Enables training on much larger datasets

- **Pretraining and Reinforcement Learning**
  - **Pretraining:** Model learns general patterns from massive datasets
  - **Reinforcement Learning:** Model is further improved using human feedback and specific examples
- **Prompt and Output Generation**
  - Users provide a prompt (input text or instructions)
  - Model generates output token by token (word by word), predicting the next most likely word
  - Output is dynamic and can vary each time, even with the same prompt
- **Probabilistic Output**
  - Model uses probabilities to choose each next word or element
  - Results can differ with each run due to randomness and temperature settings
- **Hallucinations and Limitations**
  - Models may generate incorrect or fabricated information ("hallucinations")
  - Quality of output depends on the quality and specificity of the prompt
- **Not Database Lookup**
  - Generative AI does not retrieve answers from a database
  - It predicts and generates responses based on learned patterns

## Identifying an Appropriate AI Model Based on Capabilities

### Define the Task or Problem

- Clearly identify what you need the AI to do (e.g., generate text, classify objects, detect items in images)
- Not every model is suitable for every task

### Common Types of AI Models and Their Use Cases

- **Generative AI Models**
  - Create new content (text, images, audio, code, video)
  - Best for tasks where you need to generate or synthesize data
- **Classification Models**
  - Assign items to predefined labels (e.g., is this a tree: yes or no)
  - Output includes probability scores for each label
- **Regression Models**
  - Predict numeric values based on trends in data (e.g., forecast sales)
  - Useful for time series and trend analysis
- **Clustering Models**
  - Group similar items together without predefined labels
  - Finds patterns and natural groupings in data
- **Computer Vision Models**
  - Analyze images to extract information (e.g., object detection, face recognition, text extraction)
- **Natural Language Processing (NLP) Models**
  - Understand and analyze human language (e.g., sentiment analysis, translation, summarization)

- **Speech Models**
  - Convert speech to text or text to speech
  - Handle audio-based tasks

## **Pre-built vs. Custom Models**

- **Pre-built Models**
  - Ready to use, require no training
  - Good for general tasks and quick deployment
- **Custom Models**
  - Trained on your own data for higher accuracy in specific domains
  - Useful when pre-built models are not sufficient

## **Input and Output Requirements**

- Choose models based on the type of input (text, image, audio) and desired output
- Some tasks may require chaining multiple models (e.g., image to text, then text to image)

## **Performance and Other Requirements**

- Consider model size, speed (latency), accuracy, and tolerance for hallucinations or creativity
- Some tasks may require models with explainability, citations, or specific safety features

## **Responsible AI Considerations**

- Factor in fairness, reliability, safety, and transparency when selecting a model
- Ensure the model aligns with ethical and legal requirements for your use case

## **Model Deployment Options and Configuration Parameters**

### **Model Deployment Overview**

- Deployment makes an AI model available for use by exposing an endpoint with authorization (API keys)
- Applications can send requests to the endpoint to use the model

### **Deployment Options**

- **Cloud-Based Service**
  - Model runs in the cloud; accessed over the internet
  - Common for scalability and ease of use
- **Managed Service**
  - Provider (e.g., Azure AI, Foundry tools) handles infrastructure, scaling, and updates
  - Simplifies deployment and maintenance
- **Custom Deployment**
  - Run models locally, on virtual machines, or in containers
  - Offers more control over performance, security, and integration

### **Inference Types**

- **Realtime Inference**
  - Immediate response to each request
  - Suitable for interactive applications

- **Batch Inference**
  - Processes large amounts of data at once, not in real time
  - Useful for periodic or large-scale tasks

## Key Deployment Considerations

- **Latency**
  - Time taken to get a response; large models may be slower
  - Smaller models or pre-generation can reduce latency
- **Scalability**
  - Ability to handle multiple users and high traffic
  - Important for production environments

## Configuration Parameters

- **Temperature**
  - Controls randomness/creativity of generative AI
  - Higher values = more creative, lower values = more predictable
- **Max Tokens**
  - Sets the maximum length of the model's output
- **Top-p**
  - Controls the diversity of possible outputs by limiting the pool of next-token choices
- **Content Filters and Safety Settings**
  - Restrict harmful or inappropriate outputs
  - Enable safety features as needed

## Monitoring and Logging

- Track model performance and detect issues after deployment
- Enable logging to monitor usage and outcomes over time

## Microsoft Foundry Model Deployment and Playground Demonstration

### Model Discovery and Deployment

- In Microsoft Foundry, you can search and filter available models (e.g., GPT-4o)
- Model cards provide details about each model, including use cases and limitations
- Deploy models with default or customized settings (e.g., token limits, version selection)
- Deployed models are accessible via endpoints for integration with applications

### Playground Features

- Playground allows you to interact with deployed models and simulate API usage
- Upload knowledge bases or files to enhance model responses
- Adjust content safety guardrails to filter outputs

### Parameter Configuration

- **Max Response (Tokens):** Sets the maximum length of the model's output
- **Temperature:** Controls randomness and creativity
  - Low temperature (e.g., 0.15): Output is predictable and less creative
  - High temperature (e.g., 1.3+): Output is more creative, but may become nonsensical if set too high
- **Top P:** Adjusts the diversity of possible outputs; higher values allow more varied responses

## Effect of Temperature on Output

- Low temperature: Produces safe, familiar, and predictable stories (e.g., similar to known plots)
- High temperature: Increases creativity, but may result in unusual or nonsensical content
- Moderate temperature: Balances creativity and coherence

## Scenarios for Common AI Workloads

### Generative AI

- Creates new content such as text, images, code, or audio
- Example: Generating product descriptions, creating images from prompts, writing code snippets

### Agentic AI

- Automates multi-step tasks and workflows
- Example: Meeting assistants that schedule, send invites, and reschedule based on responses

### Natural Language Processing (NLP) / Text Analysis

- Extracts meaning and insights from written text
- Techniques include:
  - **Keyword Extraction:** Identifies important words or phrases
  - **Entity Detection:** Finds names, places, dates, and labels them
  - **Sentiment Analysis:** Determines if text is positive, negative, or neutral
  - **Summarization:** Condenses long text into shorter summaries

### Speech AI

- **Speech Recognition (Speech-to-Text):** Converts spoken language into written text
  - Supports multiple languages and accents
  - Used for transcription, voice assistants, and real-time translation
- **Speech Synthesis (Text-to-Speech):** Converts text into natural-sounding audio
  - Can adjust tone, emotion, and even mimic specific voices
  - Some systems supports **neural voices**, which uses deep learning to produce more realistic speech

### Computer Vision

- Analyzes images and videos to extract information
- Features include:
  - Object detection, facial recognition, spatial analysis, and OCR (extracting text from images)
  - Example: Counting people in a video, identifying faces, reading text from signs

### Image Generation

- Creates new images from text prompts or modifies existing images
- Supports various styles, sizes, and quality levels
- Example: Generating photorealistic or artistic images based on user instructions

### Information Extraction from Text, Images, Audio, and Video

- **Text:** Use NLP for keywords, entities, sentiment, and summarization
- **Images:** Use OCR for extracting text, or computer vision for objects and faces
- **Audio:** Use speech recognition to convert audio to text, then apply text analysis
- **Video:** Extract frames for image analysis and audio for speech recognition
- **Documents:** Use AI document intelligence to extract structured data from forms, invoices, receipts, and IDs

**Other Considerations**

- Cost, latency, and privacy requirements may affect model and workload choices
- Often, multiple AI services are combined to handle complex scenarios (e.g., transcribing a meeting, summarizing the transcript, and extracting action items)